Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/31/2020
PI Name:	Allen, Jeffrey S.		
Project Title:	Testing the Equivalence of Evaporation and Condensation Coefficients Using the Constrained Vapor Bubble (CVB) Data from ISS Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	49931	**Congressional District:**	1
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	10/30/2018	**End Date:**	10/29/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/29/2021 per NSSC information (Ed., 4/22/21)		
Key Personnel Changes/Previous PI:	NOTE (Ed., Nov 2018): This work is result of proposal by graduate student Kishan Bellur, Ph.D. candidate at Michigan Technological University, who will be performing the work with advisor PI Dr. Jeffrey Allen. August 2020 report: Wonyoung Choi (studying at Case Western Reserve University) worked on this project as a visiting undergraduate researcher during the summer of 2019. Tasks included image analysis and preliminary ANSYS modeling and UDF development. August 2019 report: Since the original proposal, Kishan Bellur has completed his PhD and is now working on this grant as a post-doc.		
COI Name (Institution):	Bellur, Kishan Ph.D. (Michigan Technological University)		
Grant/Contract No.:	80NSSC19K0160		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Phase change is ubiquitous and kinetic theory is widely used to describe liquid-vapor phase change, but accurate values of evaporation coefficient and condensation coefficients are still largely unknown. For water alone, decades of research has yielded phase change coefficients that span several orders of magnitude. Evaporation and condensation coefficients are essential in analyzing liquid-vapor phase change processes. Several researchers make two primary assumptions to simplify the analysis: (1) the evaporation coefficient is equal to the condensation coefficient and (2) the liquid-vapor interface temperature is equal to the vapor temperature. Both assumptions have been shown to introduce errors. A new methodology to determine the evaporation and condensation coefficients is proposed using the Constrained Vapor Bubble (CVB) experimental data. Interferometric image analysis of the CVB data enables the reconstruction of the liquid-vapor interface to account for enhanced evaporation at the contact line. The thermal data from the experiments allows for the determination of the outer wall heat transfer characteristics and the net heat transfer to the vapor bubble. The fact that the CVB cell is a closed system allows for additional conservation equations such that the common assumptions stated above are not made. The CVB configuration along with the interferometric images, pressure, and thermal data allows for the investigation of common assumptions made in kinetic theory and for the first time, determine both the evaporation and condensation coefficients from a coupled analysis. The methodology proposed is not limited to the fluid and is applicable to any pure wetting/partially wetting fluid in the CVB setup. In addition to the development of a new methodology, the proposed work has the potential to address fundamental questions regarding commonly made assumptions in the kinetic theory of phase change. The results of the proposed research will have far reaching implications beyond space technology to applications including aerosol transport in the atmosphere, atmospheric science, and climate, micro- and nano-scale thermal transport in microelectromechanical systems (MEMS) applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Evaporation and condensation is ubiquitous, but a fundamental understanding is still lacking. This is in part due to the lack of knowledge of phase change coefficients commonly used in kinetic theory of gases. For closure and simplicity, several simplifying assumptions are generally made even though they are known to be suspect. The current work utilizes high quality data from International Space Station (ISS) experiments to test these simplifying assumptions. In addition, a new model of liquid-vapor phase change has been developed using commercial computational fluid dynamics (CFD) software in CFD. A novel “active surface” technique has been added to this software. The results of the proposed research will have impact beyond space technology to applications such as aerosol transport in the atmosphere, atmospheric science, and climate science, as well as micro and nano-scale thermal transport in microelectromechanical systems (MEMS) applications.
Task Progress:	Phase 1 involves three dimensional reconstruction of the liquid-vapor interface shape in the Constrained Vapor Bubble (CVB) experiment. Phase 1 also includes a comprehensive thermal analysis of the CVB experiment. Phase 2 is directed at modeling liquid-vapor phase change and associated transport processes; building on the shape and thermal data from phase 1. Using customized kinetic theory based expressions, the unique values of the evaporation and condensation coefficients are determined iteratively. Phase 1: Last year the three dimensional reconstruction of the interface was demonstrated using interferometric image analysis. Liquid film profiles that were extracted from the interferometric images were to be truncated at a nominal liquid film thickness of approximately 10 microns. Building on that progress, film profiles have now been extrapolated to the bulk liquid region of the CVB test cell. The interferometric film profile provides the starting conditions for the thermal modeling effort: film thickness, location, and slope. The resulting profile of the liquid-vapor interface spans the nano-scale adsorbed film to the line of symmetries on both ends of the CVB testcell. This allows for a complete reconstruction of an axial slice. A complete three dimensional reconstruction of the entire liquid-vapor interface is now possible. Thermal analysis required access to the original CVB temperature data that was not available on the Physical Sciences Informatics (PSI) database. The team obtained the necessary data directly from the original Principal Investigator (PI) of the experiment (Prof. Plawsky at Rensselaer Polytechnic Institute). The thermal analysis of the temperature data enabled the determination of the net heat added to the vapor at various heater power, and the amount of heat lost to the surroundings. The thermal analysis results will be uploaded to the PSI database at the end of the project. Currently, both the image processing and the thermal analysis are complete for one heater power of the 30 mm CVB test cell. Another issue is that the PSI database does not include images of the cold end of the CVB test cell. The research team is developing an estimated interface shape in conjunction with Prof. Plawsky’s team to address this issue. The 3D reconstruction data and thermal analysis results once completed for the other cells/heater settings will be uploaded to the PSI database at the end of the project. Phase 2: This effort is focused on development and implementation of a new methodology to determine both the evaporation and condensation coefficients. A customized kinetic theory formulation has been developed to predict the phase change at any location along the liquid-vapor interface based on the local liquid film thickness, curvature, and temperature. Liquid film thickness and curvature are readily available from the Phase 1 efforts, but liquid-vapor interfacial temperature is implicitly coupled to the phase change flux. A conjugate computational fluid dynamics (CFD) model has been developed in a commercial CFD softtware (ANSYS FLUENT). Non-uniform phase change flux along the liquid-vapor interface is modeled through a User Defined Function (UDF). This UDF identifies a region of cells near the interface as the “active region” and implements a custom mass and heat source/sink based on the customized kinetic theory formulation. The model is built such that thermocapillary effects included and this presents a unique opportunity to map the internal convective liquid flow to identify stagnation points on the liquid-vapor interface. A comprehensive analysis is forthcoming. This is a novel approach to model liquid-vapor phase change and a new technology report will be submitted. Progress on this research has been impacted by COVID restrictions at Michigan Tech. We have only recently been able to return to offices and labs. We have had issues with remote access to the primary workstation located in the on-campus research lab. It required rebooting several times and there were delays while we petitioned for building access to perform the reboot. This significantly impacted model development. We have now moved the primary workstation off-site to expedite the computational model development in anticipation of a return to restricted access.
Bibliography Type:	Description: (Last Updated: 05/04/2026)

Abstracts for Journals and Proceedings	Bellur K, Allen JS. "Multiscale modeling of liquid-vapor phase change based on experimental data." Poster presentation. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20 – 23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20 – 23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Bellur K, Allen JS. "Thermo-Mechanical Phase Change Stability of Liquid-Vapor Meniscus." Oral presentation. ASME 2019 Summer Heat Transfer Conference, AIChE Symposium in Honor of Prof. Peter C. Wayner, Jr. II, Bellevue, WA, July 14 – 17, 2019. Abstracts. ASME 2019 Summer Heat Transfer Conference, Bellevue, WA, July 14-17, 2019. , Jul-2019
Abstracts for Journals and Proceedings	Bellur K, Médici EF, Hermanson J, Hussey DS, Choi CK, Allen JS. "Mass Accommodation Coefficients of Cryogenic Propellants." Poster & Oral Presentation, 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31 – November 03, 2018. Abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31 – November 03, 2018. , Nov-2018
Awards	Bellur K, Médici EF, Hermanson J, Hussey DS, Choi CK, Allen JS. "Best Poster Presentation Award for "Mass Accommodation Coefficients of Cryogenic Propellants." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31 – November 03, 2018." Nov-2018